Indian Institute of Technology, Guwahati
Guwahati, INDIA 781 039

Department of Chemistry

Date: 09 August 2023; 7:55 a.m.

CH101

Tutorial 1; Physical Chemistry

1. The radiation power per unit area per unit wavelength is defined as: $\frac{dR}{d\lambda} = \frac{c}{4}\rho(\lambda)$

Find the expression for the total power per unit area and relate that to Stefan-Boltzmann law.

Assume: $\int_0^\infty \frac{(x)^3}{(e^x - 1)} dx = \frac{\pi^4}{15}$ and $\rho(\lambda) = \frac{8\pi hc}{\lambda^5} \frac{1}{(e^{\frac{hc}{\lambda kT}} - 1)}$

2. The amount of energy delivered to the Earth by the SUN is equal to the $\frac{dR}{d\lambda} = \frac{c}{4}\rho(\lambda)$ **energy lost to space from the Earth by Blackbody (IR) radiation. Otherwise, the Earth's temperature would continually rise (or fall). Solar radiative flux at the Earth** $S_o = \frac{\sigma T_{Sun}^4 \times r_{Sun}^2}{r_{s-e}^2}$**. This is the inverse squared distance law. S_o is solar constant for earth. It is determined from the flux at the surface of the Sun and by the distance between Earth (r_{s-e} = 1.5 x 10^{11} m) and the Sun's radius, r_{Sun}=2.3 x 10^9 m. S_o=1368 wm^{-2}. Calculate the temperature of the Earth. Assume Albedo (A) is 0.33. Albedo (A) =fraction of energy reflected away by clouds, aerosols, and the surface.**

3. Based on Planck's radiation distribution law Einstein derived the expression for the heat capacity of solid as written below. Derive the expression that would be for heat capacity at high temperatures ($h\upsilon << k_B T$).

$$c_v = \left[\frac{dU}{dT}\right]_v = \frac{3N_A k_B \left(\frac{h\upsilon}{k_B T}\right)^2 e^{h\upsilon/k_B T}}{\left(e^{h\upsilon/k_B T} - 1\right)^2}$$

4. In electron microscopy electrons are used to image objects at a very high resolution. This is done by impinging electrons at high velocity on the target substrate. The resolution limit is determined by the de Broglie wavelength ($\lambda = \frac{h}{p}$) of electrons. λ, h and p are wavelength, Planck's constant and linear momentum of the electron, respectively. What would be the best resolution achieved using a 300 kV transmission electron microscope (TEM)?